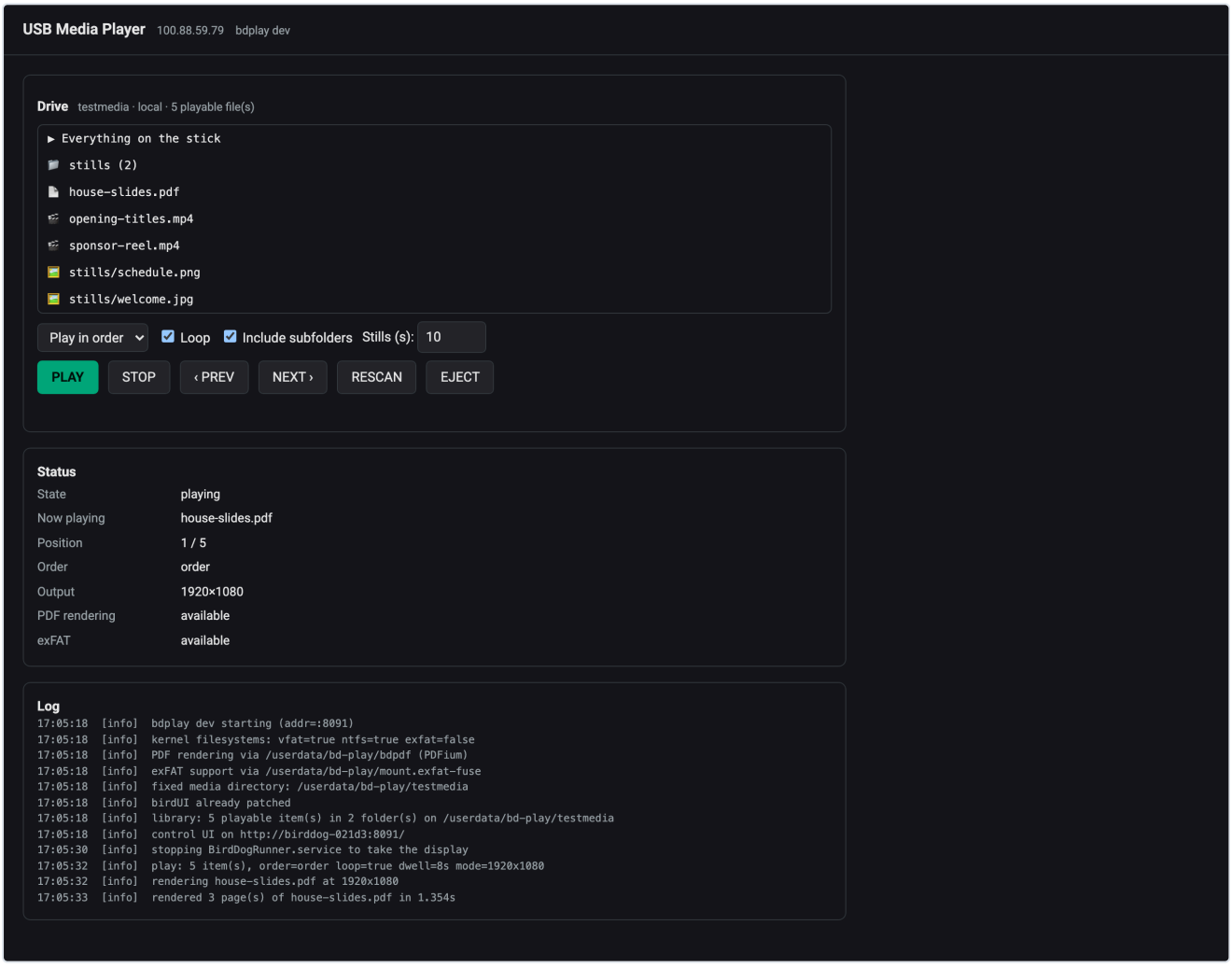
bdplay user guide

bdplay adds a **USB** entry to the BirdDog PLAY's source dropdown and **plays video, stills and PDFs from a USB stick straight out of HDMI** — hardware decoded, in order or shuffled, looping until you stop it.

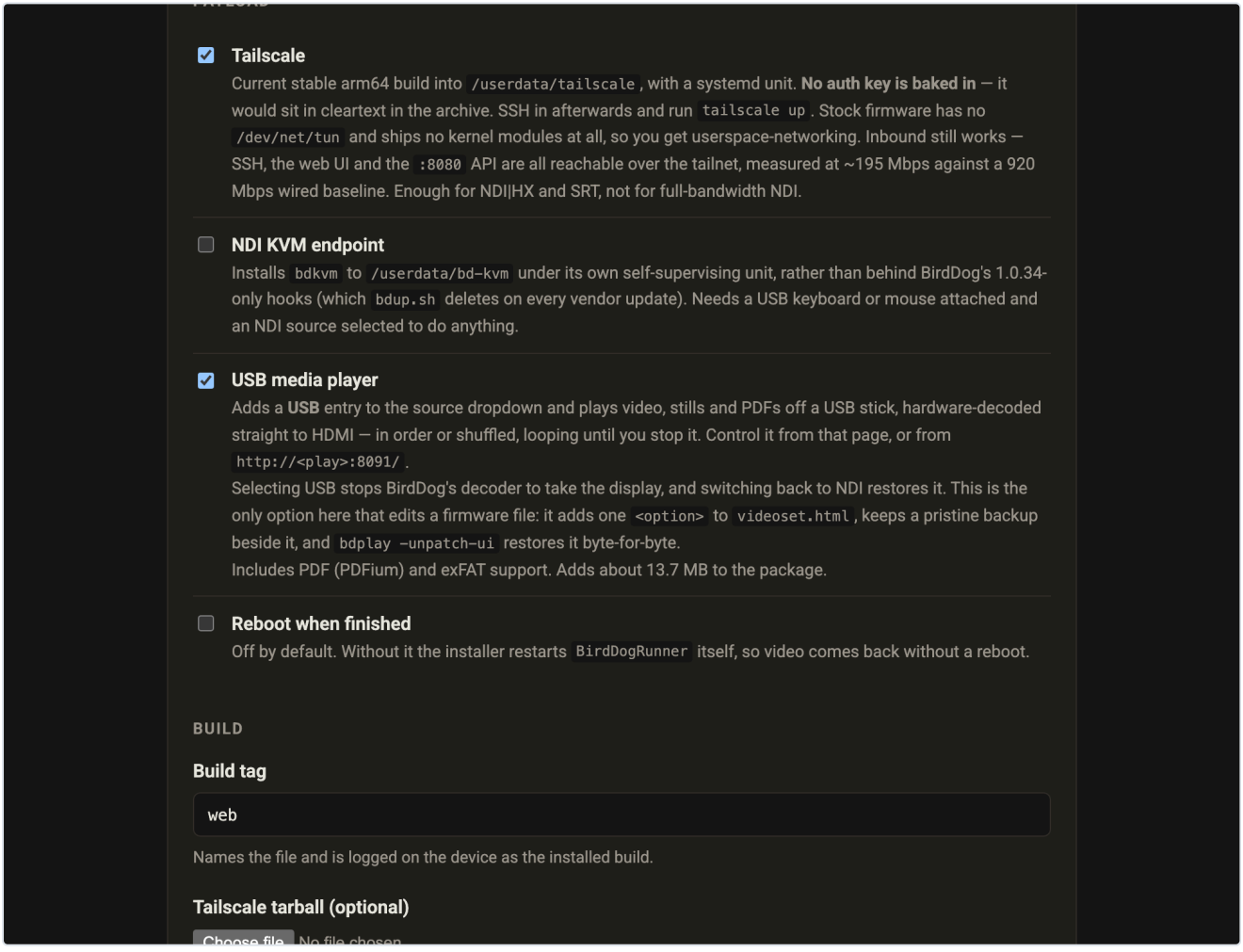
The PLAY is an NDI/SRT *decoder*: it has no notion of local media. This turns one into a signage player **without giving up anything** — select NDI again and the stock decode path comes straight back.



> **Before you rely on this:** every claim here was **measured on a real BirdDog PLAY** — hardware video decode, HDMI audio, stills, PDF rendering, the exFAT mount and the display handover with BirdDog's own decoder are all confirmed working on hardware.
>
> That is **one unit on one firmware version.** It has not been tested across the range of PLAY and Pod firmwares in the field, and **USB hotplug with a physical stick is still unverified** — testing used a loop device and internal storage.
>
> This codebase was created with AI assistance, directed and reviewed by a human author.

Installing

The easy way is a checkbox in the PLAY Patcher, which assembles the firmware package in your browser — no toolchain, nothing to build.

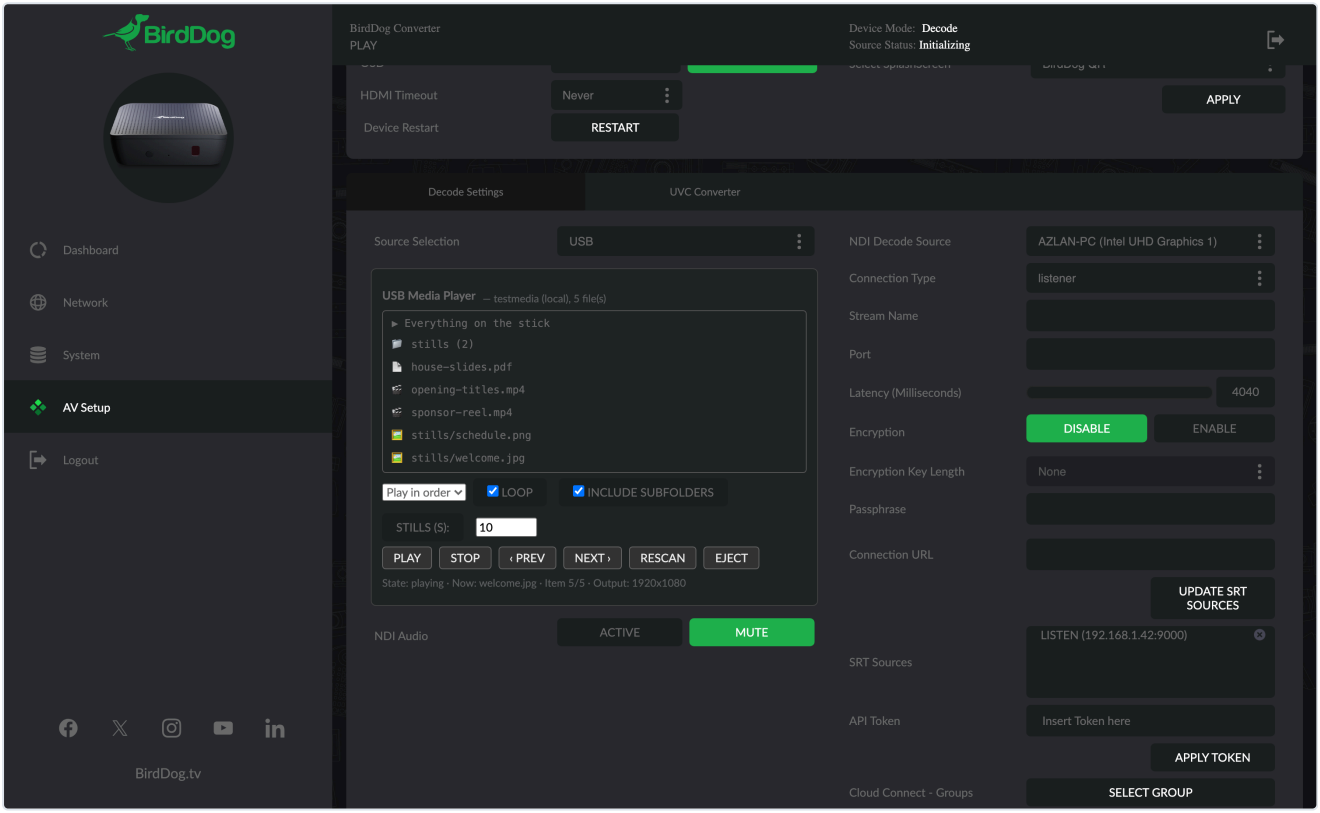


To install by hand, copy the binary and its service unit onto the unit over SSH (**port 9031** on stock BirdDog firmware) and enable the service.

On start it patches the video settings page to add the USB option. `bdplay -unpatch-ui` removes it again — the stock page is untouched apart from one added option.

Using it

Open the PLAY's own web UI, go to the video settings page, and pick **USB** from Source Selection. A panel appears with the stick's contents.



There is also a standalone control page on port 8091, which additionally shows the log — **use that if the injected panel does not appear**, and use it first when something will not play, because the log answers most of those questions.

Everything is drivable from the API too, so a Companion button or a cron job works.

From the TV's remote

If the TV supports HDMI-CEC (Samsung call it Anynet+, LG SimpLink, Sony Bravia Sync — it is usually on by default), the PLAY's own on-screen source list is already driven by the TV's remote. bdplay adds a **USB-Player** entry to that list: press **Left/Right** to reach the SRT page, **Up/Down** to it and **OK**. A file browser for the stick appears on the TV, driven by the same remote:

Key	Does
Up / Down	move through the list
OK	open a folder, or play the file
Left, **Exit / Return**	back up a folder — and at the top, back to BirdDog
Play (or the green key)	play everything in this folder, looping
Stop	while playing: back to the list — in the list: back to BirdDog
‹‹ / ›› (or Left / Right while playing)	previous / next item
Red	straight back to BirdDog
Blue / **Yellow**	shuffle / loop on and off

A single file plays once and returns to the list; a picture stays up until you press a key. The same browser can be opened from the control page on port 8091 (**OPEN BROWSER ON HDMI**), which also has the remote's keys as buttons — handy when the TV forwards only some of them, which varies by make. Arrows and OK work on every CEC TV; the transport and colour keys on most.

What it plays

Type	Formats	How
Video	mp4, mov, mkv, avi, ts, m2ts, mpg, webm, wmv	Hardware H.264/HEVC/VP8/VP9 — **~11% of one core at 1080p**
Stills	jpg, png, bmp, gif, webp, tif	Scaled to the panel with aspect preserved, held for a configurable dwell
PDF	pdf	Pages pre-rendered to images, then played as stills — 3 pages at 1080p in ~1.3 s
Office	pptx, docx, …	**Not rendered.** Listed with "export to PDF" rather than silently ignored

PowerPoint is deliberately out of scope. Rendering it on-device would mean either a full office suite — which will not fit or perform on 2 GB and four small cores — or a partial renderer that misdraws SmartArt, charts and embedded fonts, **looking broken in exactly the situations where a deck matters.** Exporting to PDF is one step and renders exactly as authored.

Playback options

- **Selection** — one file, one folder (optionally including subfolders), or the whole stick.
- **Order** — as listed (**natural sort, so `slide2` precedes `slide10`**) or random.
- **Loop** — on by default.
- **Dwell** — how long each still or PDF page holds the screen.

Random reshuffles the whole set each cycle rather than picking independently, so every file plays once per cycle instead of some repeating while others starve — and **the reshuffle never puts the item that just played at the front of the next cycle.**

If something will not play

Symptom	Cause
The USB option does not appear	The UI patch did not land. Use the standalone control page on port 8091.
A file is listed but will not play	Read the log on that page — it answers most of these.
A PowerPoint file is listed and does nothing	Deliberate. Export it to PDF.
The stick was not picked up	Hotplug with a physical stick is the one path still unverified. Re-seat it, or reboot with it in.
NDI stopped when I selected USB	Correct — the stock decoder hands over the display. Select NDI again and it comes back.
The TV remote does nothing in the browser	The TV is not forwarding keys: check its CEC setting, and that the PLAY is the input it is showing. Use the buttons on the control page meanwhile.
USB-Player is not in the PLAY's on-screen list	It is on the SRT page of that list (Left/Right), and only while the bd-play service runs.
